Sixth Semester

Math 4643 **Probability and Statistics II** **Credit: 3.00**

Review of Probability, Random Vectors; Stochastic processes: Stochastic process and their classifications, Bernoulli Process and Poisson Process, and their properties, Discrete-time and continuous-time Markov Process, stationary probabilities and balance equation, Introduction to queuing theory, Hypothesis testing - Test concerning the mean and variance of normal population, Regression and correlation, Analysis of variance - two factor analysis of variance and two-way analysis of variance with interaction, Goodness of fit test with specified and unspecified parameters, Non parametric hypothesis test- sign test.

Recommended Texts:

1. S. M. Ross, *Introduction to Probability Models*, 13th ed. Academic Press, 2023
2. S. M. Ross, *Introduction to Probability and Statistics for Engineers and Scientists*, 5th ed. Academic Press, 2014
3. D. C. Montgomery and G. C. Runger, *Applied Statistics and Probability for Engineers*, 6th ed. Wiley, 2023

CSE 4617 **Artificial Intelligence** **Credit: 3.00**

Survey of concepts in artificial intelligence. Knowledge representation, search and control techniques. All machines and features of the LISP and PROLOG languages.

Problem representation: search, inference and learning in intelligent systems; systems for general problems solving, game playing, expert consultation, concept formation and natural language procession: recognition, understanding and translation. Case Study on Expert Systems.

Recommended Texts:

1. S. J. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Pearson, 2021

CSE 4621 **Microprocessor and Interfacing** **Credit: 3.00**

Microprocessor and Assembly Language, Microprocessors and Microcomputers, Evaluation of Microprocessors Applications, Intel 8086 Microprocessor: internal architecture, register structure, programming model, addressing Modes, instruction set, Assembly language programming, Intel x86 and x64 architecture - overview. Interrupts, address space partitioning, A-to-D and D-to-A converters and some related chips. Interfacing ICs of I/O devices - I/O ports, Programmable peripheral interface, DMA controller, interrupt controller, communication Interface, interval time, etc. IEEE 488 and other buses, interfacing with microcomputer. Interfacing I/O devices - floppy

disk, hard disk, tape, CD-ROM & other optical memory, keyboard, mouse, monitor, plotter, scanner, etc.

Microprocessor in Scientific Instruments and other applications - Display, Protective Relays, Measurements of Electrical quantities, Temperature monitoring system, water level indicator, motor speed controller, Traffic light Controller, etc. Microprocessor based interface design.

Recommended Texts:

1. D. V. Hall, *Microprocessors and Interfacing*, 2nd ed. McGraw-Hill, 2005
2. Y. Y. Ytha and C. Marut, *Assembly Language Programming and Organization of the IBM PC*, 1st ed. McGraw-Hill/Irwin, 1992
3. B. B. Brey, *The Intel Microprocessors 8086/8088, 80186/80188, 80286, 80386, 80486, Pentium, and Pentium Pro Processor Architecture, Programming, and Interfacing*, 6th ed. Pearson, 2002
4. P. Marwedel, *Embedded System Design: Foundations of Cyber-Physical Systems, and the Internet of Things*, 4th ed., N. D. Dutt and G. Martin, Eds. Springer, 2021

SWE 4601 **Software Design and Architectures** **Credit: 3.00**

Introduction, Design Concepts, Review of UML, Object Oriented Analysis and Design, Study of the Design of some software: Library, Text Editor, Compiler, E-Commerce Site, Mobile Application, Design Patterns, Design Principles, User Interface Design: The Golden Rules, User Interface Analysis and Design, Interface Analysis, Interface Design Steps, Web App Interface Design, Design Evaluation.

Software Architecture, Architectural Views, Architectural Styles: Object Oriented Architecture, Data Driven Architecture, Client Server Architecture, Service Oriented Addison-Wesley Architecture, Component Based Architecture, Web Architecture, Mobile Software Architecture, Connectors, Middleware, Message Queue, Web Service, XML, Non-Functional Requirements, Architectural Trade-offs, Software Redesign and Architectural Migration.

Recommended Texts:

1. H. Gomaa, *Software Modeling and Design: UML, Use Cases, Patterns, and Software Architectures*, 1st ed. Cambridge University Press, 2011
2. E. Evans, *Domain-Driven Design: Tackling Complexity in the Heart of Software*, 1st ed. Addison-Wesley Professional, 2003
3. R. S. Pressman and B. R. Maxim, *Software Engineering: A Practitioner's Approach*, 9th ed. McGraw-Hill, 2019